

Name: DARSHAN D

Reg No: 192524117

DSA0602 Data Handling and Visualization

CO3 AT2 — Data Analysis Project

Project Title: Data Analysis & Dashboard using the Bank Marketing dataset

Scenario: You have been engaged as a data analytics consultant by a private retail bank. The Head of Direct Marketing has commissioned a rigorous, evidence-based study using the publicly available “Bank Marketing dataset” (source: UCI) with the goal of evaluating the effectiveness of a term-deposit telemarketing campaign to redesign next quarter's outreach. Rather than another static slide deck, the Head of Direct Marketing has specifically asked for a single interactive dashboard that a non-technical decision-maker can explore independently.

Dataset: Download and use the “Bank Marketing dataset” dataset from UCI (or an equivalent up-to-date mirror of it). Typical fields include: age, job type, marital status, contact method, campaign call count, and subscription outcome. If the exact published version has fewer than 300 usable rows after cleaning, you may supplement it with a clearly documented simulated extension — but the core analysis must be performed on the real, downloaded data.

Required Analyses & Visual Components

Required Chart	What it Must Show
1. Visualizing Proportions: Pie Charts	A pie chart showing the proportional breakdown of a key categorical variable, with percentage labels on each slice.
2. Sunburst Charts	A sunburst chart of a hierarchical breakdown (ideally the same hierarchy as your treemap), allowing drill-down from an outer to an inner category.
3. Avoiding Misleading Proportions	One deliberately misleading proportion chart built from your dataset, and a corrected version, with a short written note on exactly what was fixed.
4. Visualizing Many	A single chart comparing the distribution of a key variable across at least 5

Distributions at Once	categories/groups present in the dataset.
5. Side-by-Side Bars	A side-by-side (dodged) bar chart comparing a metric across categories, split further by a second grouping variable.
6. Parallel Sets	A parallel sets / parallel categories diagram tracing how records flow across at least two categorical dimensions in the dataset.

Deliverables & Tasks

(a) Data Preparation & Documentation: Download and clean the dataset; handle missing values and clear outliers; document every field you use, any transformations applied, and any assumptions made.

(a) Data Preparation & Documentation

Dataset

- **Dataset Name:** Bank Marketing Dataset
- **Source:** UCI Machine Learning Repository
- **Total Records:** 45,211
- **Total Attributes:** 17

Fields Used

Field	Description
age	Customer's age
job	Type of occupation
marital	Marital status
education	Education level
contact	Communication method (Cellular/Telephone)
campaign	Number of contacts during the campaign
month	Last contact month
poutcome	Outcome of previous campaign

Field	Description
y	Whether the customer subscribed to a term deposit (Yes/No)

Data Cleaning

- Downloaded the dataset from the UCI repository.
- Removed duplicate records.
- Checked for missing values and handled them appropriately.
- Treated "unknown" as a separate category where required.
- Converted categorical variables into appropriate formats.
- Removed extreme outliers from **age** and **campaign** using the IQR method.
- Verified the cleaned dataset before analysis.

Transformations

- Converted the target variable (**y**) into Yes/No categories.
- Grouped ages into age groups (18–30, 31–45, 46–60, Above 60).
- Calculated percentages for categorical variables.
- Aggregated customer counts for visualization.

Assumptions

- Unknown values were retained as a valid category.
- Outliers beyond the IQR range were removed only from numerical variables.
- The dataset accurately represents one telemarketing campaign.

(b) Analytical Charts (25 Marks)

1. Pie Chart – Visualizing Proportions

Chart

Subscription Outcome (Yes vs No)

Reason

A pie chart effectively shows the percentage distribution of customers who subscribed and those who did not.

Interpretation

The majority of customers did not subscribe to the term deposit, while only a smaller proportion accepted the offer.

2. Sunburst Chart

Hierarchy

Job → Marital Status → Subscription Outcome

Reason

The Sunburst chart displays hierarchical relationships and allows users to drill down through different customer categories.

Interpretation

Management and retired customers showed relatively higher subscription rates, while married customers formed the largest customer group.

3. Misleading Proportion Chart

Misleading Chart

A pie chart was created using only customers who subscribed, making each category appear larger than it actually is.

Corrected Chart

The corrected chart includes all customers in the dataset and calculates percentages based on the total number of customers.

What was Fixed?

- Used the complete dataset instead of a subset.
- Calculated percentages correctly.
- Added an accurate chart title.
- Included proper labels and legends.

4. Visualizing Many Distributions at Once

Chart

Violin Plot

Variable

Age

Groups

- Admin
- Technician
- Management
- Blue-collar
- Services

Reason

A violin plot compares the distribution, density, spread, and median of age across multiple job categories.

Interpretation

Management and technician jobs generally have higher age distributions, while service workers tend to be younger.

5. Side-by-Side Bar Chart

Variables

- X-axis: Contact Method
- Y-axis: Number of Customers
- Group: Subscription Outcome

Reason

A side-by-side bar chart enables direct comparison of subscription outcomes across different contact methods.

Interpretation

Customers contacted through cellular communication had a higher number of successful subscriptions compared to telephone communication.

6. Parallel Sets Diagram

Dimensions

Job → Education → Subscription Outcome

Reason

The parallel sets diagram illustrates how customers move across multiple categorical variables and their final subscription outcome.

Interpretation

Customers with higher education working in management occupations contributed more to successful subscriptions.

(c) Interactive Dashboard Design

Tool Used

Tableau

Dashboard Components

- Pie Chart
- Sunburst Chart
- Violin Plot
- Side-by-Side Bar Chart
- Parallel Sets Diagram
- Misleading vs Corrected Pie Chart

Interactive Filters

- Age Group
- Job
- Marital Status
- Contact Method
- Month

Interactivity

- Selecting a filter updates all charts simultaneously.
- Hovering over charts displays detailed information.
- Users can drill down into the Sunburst chart.
- Clicking on one chart highlights related information in the other charts.

Dashboard Design

The dashboard is organized into clearly labelled sections with consistent colors, legends, and explanatory notes to ensure easy understanding for non-technical users.

(d) Insights & Recommendations

Key Insights

Insight 1

Most customers did not subscribe to the term deposit, indicating that the overall campaign conversion rate was low.

Insight 2

Cellular communication resulted in more successful subscriptions than telephone communication.

Insight 3

Management and retired customers showed higher subscription rates compared to other occupations.

Insight 4

Customers who received a large number of campaign calls were less likely to subscribe, suggesting that repeated contact reduced campaign effectiveness.

Insight 5

Higher education levels were associated with better subscription outcomes.

(e) Presentation & Usability

The dashboard was designed to be simple and user-friendly for non-technical decision-makers.

- A clear dashboard title is provided.
- Every chart includes appropriate axis labels and legends.
- A consistent color palette is used throughout the dashboard.
- Interactive filters allow users to explore different customer groups.
- Hover tooltips provide additional details for each chart.
- Short explanatory notes accompany each visualization.
- The dashboard layout is clean, well-organized, and easy to navigate.

Conclusion

The analysis shows that customer occupation, education level, communication method, and campaign strategy significantly influence term deposit subscriptions. Customers contacted

through **cellular communication**, particularly those in **management** or **retired** occupations and with **higher education levels**, were more likely to subscribe. Reducing repeated calls and targeting these high-potential customer groups can improve the effectiveness of future telemarketing campaigns and increase the bank's subscription rate.